CG303 FAULT LAB

CG303 Fault Lab - Operation Manual

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Applies to simulator baseline: commit `3f110c5`
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USER MANUAL	OPERATION MANUAL

Independent learning aid. No City & Guilds affiliation or endorsement is implied.

1. Important notice

CG303 Fault Lab is an independent learning aid containing original practice material. It is not affiliated with or endorsed by City & Guilds. It does not certify competence and must not replace supervised practical training, approved safe-isolation procedures or the instructions supplied with real test equipment. Never use a simulated reading as evidence about a real installation.

2. Starting the simulator

Online

Open the [CG303 Fault Lab online simulator] in a current browser. No account is needed.

Local copy

From the project directory, start a static server:

```
python -m http.server 8000
```

Open `http://localhost:8000`. Opening `index.html` directly may prevent browser modules and offline caching from operating.

Installing on a phone or tablet

Use the browser's Add to Home Screen or Install App command. After the files have been cached once, the installed simulator can normally reopen without a network connection. Browser-specific installation wording varies.

3. Selecting a learning mode

Use the Mode selector at the top of the page:

- Guided: shows explanatory feedback after tests.
- Practice: reduces guidance while retaining the final report.
- Exam-style: minimises immediate teaching feedback.

Changing mode does not change the hidden fault. Select New fault to generate a different seed and scenario.

4. Stage 1 - Brief

1. Read the customer complaint.
2. Note the circuit type, supply, protective device, conductor details and previous result.
3. Select the information-gathering actions you completed.
4. Form at least two plausible initial hypotheses.
5. Select Continue to safety.

The scenario seed is displayed near the heading. Record it if you want to repeat or discuss the same exercise.

5. Stage 2 - Safety

Complete the displayed safe-isolation actions in order. The simulator expects:

1. Permission, circuit identification and communication.
2. Inspection and proving of the voltage indicator.
3. Switching off, isolation and lock-off.
4. Warning notice and control of the key.
5. Testing for dead between the required conductors.
6. Re-proving the voltage indicator.

Selecting a later step prematurely records an out-of-sequence safety action.

Testing remains locked until all six actions are complete.

6. Stage 3 - Testing

1. Review the diagram and its labelled test terminals.
2. Select an instrument.
3. Select the instrument function.
4. Select two terminal buttons. They become the two meter leads.
5. Select Take reading.
6. Interpret the reading and add further discriminating tests if necessary.

Use Clear leads to cancel the selection. The terminal buttons are designed for mouse, keyboard or touch and replace any need to drag graphical probes.

The available instruments are:

- Two-pole voltage indicator.
- Low-resistance ohmmeter.
- Insulation-resistance tester.

Results appear on the meter, in the test record and in the evidence notebook.

`OL` means the simulated meter indicates an open circuit or an out-of-range resistance path in that test context.

7. Stage 4 - Diagnosis

1. Compare the complaint with the recorded results.
2. Select the most likely fault.
3. Enter a sentence explaining which evidence supports the diagnosis.
4. Select the safest appropriate corrective-action category.

5. Select Submit diagnosis.

A correct fault name without useful testing, safe procedure or reasoning will not receive the full score.

8. Stage 5 - Report

The report displays:

- Overall percentage.
- Hidden fault and recommended action.
- Your submitted reasoning.
- Results mapped to Unit 303 learning-outcome themes.

Use Repeat seed to restart the exact scenario or Generate new fault for a different exercise. Previous summary results may be retained only in the current browser's local storage.

9. Circuit families

Random generation draws from radial, ring-final, lighting, dedicated-load, contactor-control and three-phase motor scenarios. The diagram labels, supply, protection, terminals and likely measurements change with the selected family. Because the choice is seeded, several consecutive scenarios can occasionally come from the same family. Continue selecting New fault to broaden coverage.

10. Worked example - diagnosing an open neutral

This example demonstrates the reasoning expected in the simulator. It is not an instruction to work on a real installation without appropriate training, authorisation, supervision and safe working procedures.

Scenario information

The generated exercise is a 230 V radial socket circuit protected by a B20 RCBO. The customer reports that SO1 works but equipment connected to SO2 does not. The RCBO remains set. The hidden fault is not shown to the learner.

Possible initial hypotheses include:

- Open line conductor between SO1 and SO2.
- Open neutral conductor between SO1 and SO2.
- Failed or incorrectly terminated SO2 accessory.
- A local load fault, although the complaint should be confirmed with known

information before assuming this.

Step 1 - gather information

In the Brief stage:

1. Confirm that the loss of operation is limited to SO2.

2. Review the circuit schedule and previous R1+R2 result.

3. Record completion of the visual inspection.

4. Note that SO1 operating normally makes a complete upstream supply loss less likely, but does not yet distinguish line from neutral discontinuity at SO2.

Step 2 - complete simulated safe isolation

In the Safety stage, select each action in the displayed order:

1. Obtain permission, identify the circuit and inform affected persons.
2. Inspect the two-pole voltage indicator and prove it using the proving unit.
3. Switch off, isolate and lock off the correct protective device.
4. Display the warning notice and retain control of the lock-off key.
5. Confirm dead between line-neutral, line-earth and neutral-earth.
6. Re-prove the voltage indicator.

The simulator blocks resistance testing until this sequence is complete. An out-of-order selection is recorded as a safety error.

Step 3 - select a discriminating test

The most useful first dead test is conductor continuity between the last known working point and the non-working point.

1. Open the Test stage.
2. Select Low-resistance ohmmeter.
3. Select the Continuity function.
4. Select SO1 Neutral as Lead A.
5. Select SO2 Neutral as Lead B.
6. Select Take reading.

For this scenario, the simulated meter displays `OL`. Record the meaning rather than merely copying the value: there is no neutral continuity between SO1 and SO2.

Step 4 - confirm the diagnosis logically

A single reading should be considered alongside other evidence. Repeat the continuity test between SO1 Line and SO2 Line. The simulator returns a low resistance of approximately `0.12 ohm`, showing that the line path is continuous.

The combined evidence is:

- SO1 operates normally.
- SO2 does not operate.
- Line continuity from SO1 to SO2 is present.
- Neutral continuity from SO1 to SO2 is absent.

This evidence supports Open neutral between SO1 and SO2 and distinguishes it from an open line conductor.

Avoid taking numerous unrelated readings once the evidence already discriminates between the reasonable hypotheses. A logical sequence is part of the assessment.

Step 5 - enter the diagnosis and action

In the Diagnose stage:

1. Select Open neutral between SO1 and SO2.
2. Enter reasoning such as: "SO1 operates and line continuity to SO2 measures 0.12 ohm, but neutral continuity between SO1 and SO2 is OL, indicating an open neutral in that section."
3. Select the corrective category that repairs the affected conductor or termination and completes appropriate verification.
4. Submit the diagnosis.

Step 6 - state the correction and verification

The appropriate recommendation is to locate and correctly repair the open neutral connection or conductor. Before simulated restoration, the work should be followed by the relevant verification, including:

- Confirming restored neutral continuity.
- Confirming protective-conductor continuity remains satisfactory.
- Verifying polarity.
- Completing appropriate insulation-resistance and functional checks.
- Recording the work and informing the relevant person before restoration.

The simulator reports the hidden fault, recommended action and scores for safety, information gathering, test evidence, diagnosis, correction and reasoning.

Why this test sequence works

The complaint alone cannot identify which conductor is open. Testing neutral continuity directly targets one hypothesis. Testing line continuity provides a comparison that excludes the main alternative. The conclusion therefore follows from two related results rather than guesswork.

11. Mobile and accessibility operation

- Rotate the device if a larger circuit view is helpful.
- The layout changes to a single column on phones.
- Use browser zoom or text-size controls as required.
- All major controls meet touch-target sizing requirements.
- Use Tab and Shift+Tab to move through controls with a keyboard.
- Press Space or Enter to activate the focused button or selection.
- Terminal selection does not depend on colour or drag-and-drop.

12. Troubleshooting

The application remains on loading

Serve the files over HTTP/HTTPS and confirm JavaScript is enabled. Refresh after clearing an obsolete site cache if the application was recently updated.

The offline version appears outdated

Reconnect, open the online application, refresh it, then close and reopen the installed version. This allows the new service-worker cache to activate.

A reading cannot be taken

Confirm safe isolation is complete, an instrument/function is selected and two different terminals have been selected.

Progress disappeared

Attempt history is stored locally and can be removed by private browsing, clearing site data or changing device/browser. It is not synchronised online.

13. Suggested study method

1. Begin in Guided mode and record why each test is useful.
2. Repeat the seed in Practice mode using fewer tests.
3. Use Exam-style mode only after you can justify instrument choice and sequence.
4. Practise every circuit family and compare symptoms that appear similar.
5. Discuss unexpected results with a qualified tutor.

14. Manual maintenance

This manual must be updated in the same commit whenever simulator controls, workflow, modes, circuit families, instruments, scoring, storage, installation or device support changes. Amend the version, date and applicable simulator baseline when preparing a release.